COMPOSITIONS VANISHING ON KERNELS AND COKERNELS VANISH IN THE HOMOTOPY CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3148-3155, immediately following the statement at 3119-3146.

Editorial note. The hypothesis diagrams commute up to homotopy in K(A); the replacement makes the relevant squares strictly commute.

Allowed standard prerequisites (editorial).

- Strict commutativity replacement. Lemma lemma-make-commute-map of this chapter, cited at proof lines 3148-3150. Replaces b and b' by homotopic maps making one square of each diagram strictly commute.

Source reference. Original lemma 014H: sources/stacks-derived/pinned-original-source.tex, source lines 2808–2808. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $0 \to A_i^\bullet \to B_i^\bullet \to C_i^\bullet \to 0$, $i = 1, 2, 3$ be termwise split exact sequences of complexes. Let $b : B_1^\bullet \to B_2^\bullet$ and $b' : B_2^\bullet \to B_3^\bullet$ be morphisms of complexes such that*

$$\begin{array}{ccccc} A_1^\bullet & \longrightarrow & B_1^\bullet & \longrightarrow & C_1^\bullet \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b}\downarrow & & {\scriptstyle 0}\downarrow \\ A_2^\bullet & \longrightarrow & B_2^\bullet & \longrightarrow & C_2^\bullet \end{array} \quad \text{and} \quad \begin{array}{ccccc} A_2^\bullet & \longrightarrow & B_2^\bullet & \longrightarrow & C_2^\bullet \\ \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle b'} & & \downarrow{\scriptstyle 0} \\ A_3^\bullet & \longrightarrow & B_3^\bullet & \longrightarrow & C_3^\bullet \end{array}$$

commute in $K(\mathcal{A})$. Then $b' \circ b = 0$ in $K(\mathcal{A})$.

Proof. By Lemma 014H we can replace b and b' by homotopic maps such that the right square of the left diagram commutes and the left square of the right diagram commutes. In other words, we have $\mathrm{Im}(b^n) \subset \mathrm{Im}(A_2^n \to B_2^n)$ and $\mathrm{Ker}((b')^n) \supset \mathrm{Im}(A_2^n \to B_2^n)$. Then $b' \circ b = 0$ as a map of complexes. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.